Respiratory syncytial virus infection in the modern era

Martin Ludlow

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Purpose of review
Respiratory syncytial virus (RSV) continues to be a major cause of severe lower respiratory tract infection in infants, young children, and older adults. In this review, changes in the epidemiology of RSV during the coronavirus disease 2019 (COVID-19) pandemic are highlighted together with the role which increased molecular surveillance efforts will have in future in assessing the efficacy of vaccines and therapeutics.

Recent findings
The introduction of nonpharmaceutical intervention (NPIs) strategies during the COVID-19 pandemic between 2020 and 2022 resulted in worldwide disruption to the epidemiology of RSV infections, especially with respect to the timing and peak case rate of annual epidemics. Increased use of whole genome sequencing along with efforts to better standardize the nomenclature of RSV strains and discrimination of RSV genotypes will support increased monitoring of relevant antigenic sites in the viral glycoproteins. Several RSV vaccine candidates based on subunit, viral vectors, nucleic acid, or live attenuated virus strategies have shown efficacy in Phase 2 or 3 clinical trials with vaccines using RSVpreF protein currently the closest to approval and use in high-risk populations. Finally, the recent approval and future use of the extended half-life human monoclonal antibody Nirsevimab will also help to alleviate the morbidity and mortality burden caused by annual epidemics of RSV infections.

Summary
The ongoing expansion and wider coordination of RSV molecular surveillance efforts via whole genome sequencing will be crucial for future monitoring of the efficacy of a new generation of vaccines and therapeutics.

Keywords
disease burden, molecular epidemiology, pathogenesis, respiratory syncytial virus, therapeutics, vaccines

INTRODUCTION

The progressive easing of legally mandated restrictions related to the coronavirus disease 2019 (COVID-19) pandemic in 2021 and 2022, resulted in a rebound in the transmission of many respiratory viruses such as influenza A virus (IAV) and respiratory syncytial virus (RSV), following an almost complete absence of verified infections in the 2020/2021 winter season [1]. This contributed to a ''tripledemic'' of SARS-CoV-2, IAV, and RSV infections in the 2022/2023 winter season [2], which led to an increased burden on healthcare systems worldwide with RSV infections being primarily responsible for the hospitalization of large numbers of pediatric patients with acute lower respiratory tract infection (LRTI). Underlying this altered epidemiology of endemic respiratory human viruses has been an increased pool of young children who have never been exposed to RSV and are thus seronegative (RSV-naive), the altered age distribution of this susceptible population and the complex dynamics influencing epidemic peaks of co-circulating respiratory viruses. The recent increased focus on RSV in the wider media has facilitated a better awareness of which population groups are particularly vulnerable to severe infection, the routes of virus transmission, and the opportunities presented by a new generation of vaccines and therapeutics. The latter may help to finally remediate the high levels of morbidity and mortality associated with annual epidemics of RSV infections. It is thus timely to assess the

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KEY POINTS

- The use of nonpharmaceutical interventions during the COVID-19 pandemic disrupted the timing and peak incidence of previously predictable annual RSV seasons.
- The expansion of whole genome sequencing of RSV from clinical samples will facilitate monitoring of variation in antigenic sites present in the F protein, which are relevant for the efficacy of vaccines and therapeutics.
- Several vaccine candidates have proven efficacious in clinical trials with subunit vaccines based on inoculation of RSV pre-F protein likely to be used first in older adults at risk of RSV-induced severe LRTI.
- The human mAb Nirsevimab with an extended half-life binds to antigenic site Ø in the pre-fusion confirmation of the RSV F protein and has been approved for use in in neonates and infants during their first RSV season.

current direction of the RSV field with respect to molecular epidemiology, pathogenesis, vaccines, and therapeutics.

IMPACT OF THE COVID-19 PANDEMIC ON RESPIRATORY SYNCYTIAL VIRUS EPIDEMIOLOGY

The timing and scale of epidemics of RSV are well characterized and typically occur annually between March and June in the Southern hemisphere and September and December in the Northern hemisphere, with longer seasons of up 10 months occurring in countries located in tropical regions [3]. However, the introduction of nonpharmaceutical intervention (NPIs) strategies during the COVID-19 pandemic resulted in abnormal circulation patterns of RSV in 2020 with an almost complete absence of RSV transmission during the expected annual epidemic peak in both the southern and northern hemisphere winter seasons [1,4▪]. The progressive easing of NPI measures in many countries worldwide in 2021 resulted in out of season surges in RSV cases, with the timing dependent on when increased social activities were resumed [3,5,6]. An altered age distribution was also evident in some regions as the numbers of laboratory-confirmed RSV cases in England in children aged between 1 and 4 years [4▪] and in Rhône Loire, France, in children aged between 3 months and 5 years [7] were higher than in typical pre-COVID-19 winter seasons. The 2022/2023 RSV season again started earlier, but the peak of cases in Denmark in week 45 was temporally closer to the typical peak observed in pre-pandemic seasons [8]. Analysis of RSV whole genome sequences (WGS) derived from samples obtained at the start of the early large surge in cases in the greater Boston area in the USA in November 2022 revealed the co-circulation of strains belonging to multiple lineages of RSV-A and RSV-B rather than a single more transmissible variant strain [9▪▪]. The fact that these strains last shared a common ancestor before the COVID-19 pandemic provides support to the attribution of out of season surges of RSV in 2021 and 2022 to reduced levels of population immunity following relaxation of NPI strategies. It is not yet understood how the transition of SARS-CoV-2 to endemicity will impact on the future timing and severity of annual RSV epidemics. However, the complex interactions resulting from co-circulation of endemic respiratory viruses is of increased interest given the increased mortality observed in hospitalized SARS-CoV-2 patients with a respiratory viral co-infection [10] and recent results showing that pre-infection of BALB/c mice or alveolar epithelial cells by RSV triggers an innate immune response, which reduces the replication of influenza virus (IAV) during a subsequent infection [11,12].

RESPIRATORY SYNCYTIAL VIRUS EPIDEMIOLOGY

The wider field of virus molecular epidemiology has advanced rapidly in recent years in both scale and sophistication due to technical improvements in next-generation sequencing (NGS) and associated data storage, standardization of protocols, and the establishment of online platforms such as GISAID (GISAID - gisaid.org) to facilitate the rapid curation and sharing of sequencing data. This enabled almost real-time monitoring during the COVID-19 pandemic of increases in the rate of transmission of new variants of SARS-CoV-2 in defined geographical locations. Similarly, whole genome sequencing (WGS) of RSV strains has rapidly increased in the past 5 years, and such sequencing activities will undoubtedly benefit from the more widespread sequencing infrastructure and technical experience, which has been acquired worldwide during the COVID-19 pandemic (Fig. 1). In addition, technical advances in the detection and sequencing SARS-CoV-2 variants in wastewater samples [13] are also being applied to the field of RSV epidemiology, which will augment the monitoring of specific clades of RSV, which is normally performed via sequencing of clinical samples obtained from RSV cases [14]. This may provide early indications for higher levels of virus transmission within communities, thus providing valuable time for pediatric departments in local hospitals to prepare for

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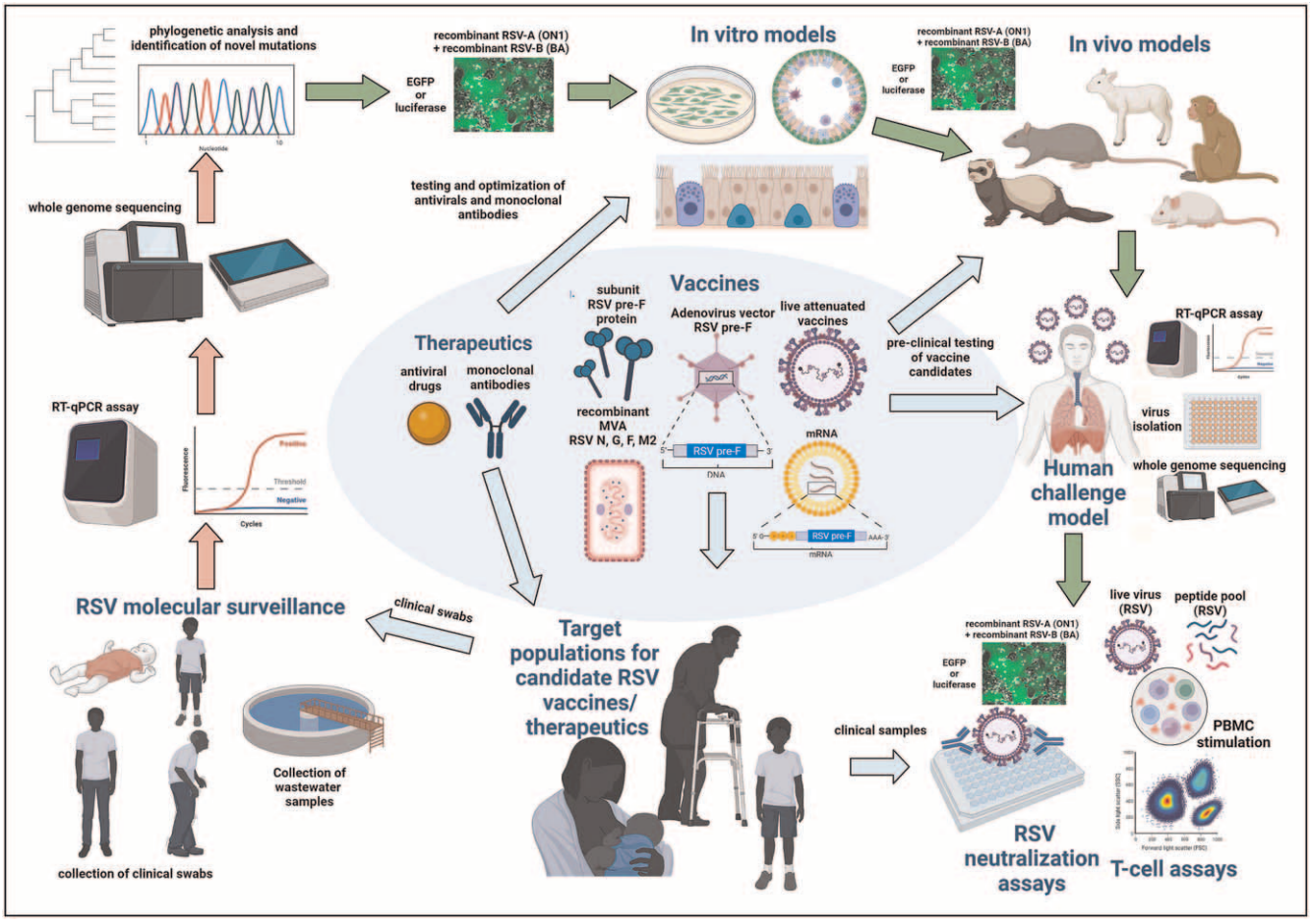


FIGURE 1. A schematic overview showing that integration between RSV molecular surveillance, in vitro and in vivo model systems and associated assays are necessary to assess the testing, validation, and efficacy of vaccines and therapeutic candidates in populations susceptible to severe RSV-induced lower respiratory tract infection. Created with BioRender.com.

additional cases of RSV-mediated severe LRTI. Recent technical improvements to WGS workflows include using amplicon-based NGS on clinical samples obtained from RSV patients [15▪,16]. This strategy uses Illumina or Oxford Nanopore platforms to sequence pooled amplicons generated via RT-PCR using primers targeted to conserved regions of RSV-A and RSV-B genomes. An alternative strategy is target capture enrichment NGS in which a probe set designed against published RSV-A and RSV-B sequences has been used to obtain complete genome sequences from clinical samples collected during an outbreak of RSV-B in adult hematologic patients [17]. This can facilitate infection control measures within hospital settings by revealing routes of virus transmission.

Analysis of RSV fusion (F) sequences remains an important complement to WGS and can reveal variation in antigenic sites (Ø, I-V), thus enabling identification of specific mutations, which might reduce the efficacy of vaccines and/or therapeutics [18,19]. Similarly, sequencing of G genes has shown the presence of mutations in the two large mucin like domains and surprisingly a single amino acid change (N178G) in the conserved central domain (CCD) of the G protein [20▪]. Sequences derived from the second hypervariable region of the G gene are also still commonly used for phylogenetic analyses to determine the evolutionary relationship of different RSV strains. However, the increasing numbers of WGS for RSV and wider organization of sequencing activities at both national and international levels (WHO's Global RSV surveillance program) have led to coordinated efforts to better standardize the nomenclature and classification and of RSV strains. This is already established for other viruses which spread via the respiratory route such as measles virus or IAV and follows the earlier re-classification by the International Committee on Taxonomy of Viruses (ICTV) in 2016 of RSV into the species *Human orthopneumovirus* within the genus *Orthopneumovirus* in the family *Pneumoviridae* [21].

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RSV strains are now recommended to be named using this method: [virus name abbreviation]/[HRSV subgroup]/[geographic identifier]/[unique sequence identifier]/[year of sampling], for example HRSV/A/USA/001/2011 [22▪]. Two attempts have recently been made to better standardize the classification of RSV genotypes. An analysis of G-ectodomain sequences by Goya *et al*. [23] resulted in the discrimination of three RSV-A genotypes (GA1-GA3) and seven RSV-B genotypes (GB1-GB7). Alternatively, a separate study has determined that WGS alignments provide the most reliable method to discriminate between RSV genotypes with 23 RSV-A genotypes and six RSV-B genotypes identified from available RSV sequences [24]. It is important that either the ICTV or the RSV field in general soon determines which form of classification should be considered best practice for future sequencing studies.

VACCINES

The disease burden of RSV infections in young children, adults and older adults has been undergoing a re-evaluation in recent years to facilitate planning of future vaccination strategies and provide better estimates of the economic costs incurred by annual RSV epidemics. A systematic review and meta-analysis has shown that RSV and influenza virus infection of older adults leads to a similar rate of hospitalization and mortality [25]. In contrast, a retrospective cohort study from Eastern Finland in 2017–2018 showed that RSV-infected patients were older and had an increased risk of hospitalization and mortality in comparison to influenza virus-infected adults [26▪▪]. The level of RSV infection in adults and older adults may also be underestimated due to limitations in diagnostic testing, a situation that could be alleviated by the collection and testing of three or more specimen types from patients [27]. The burden of RSV-infection in healthy term infants has also been recently assessed in a prospective birth cohort study performed in five locations in Europe [28]. Diagnostic testing revealed a 26.2% incidence of RSV infection within the cohort with 14.1% of children having a medically attended RSV infection and a 1.8% incidence of RSV-associated hospitalization, confirming the need for active immunization of pregnant women or healthy term-born infants [28].

The long-term preclinical development of different RSV vaccination strategies has resulted in several different subunit, recombinant viral vector, nucleic acid, particle based, chimeric and live attenuated vaccines, which have displayed varying levels of efficacy in phase 1–3 clinical trials in humans (Fig. 1) [29]. Currently, these vaccines are aimed for use in older individuals (>55 years of age) but in the longer term will also be targeted toward pregnant women to enhance RSV-specific maternal antibodies in infants (0–6 months) and in young children (6 months–5 years of age). Many of these vaccine candidates rely on expression of RSV F protein in the prefusion conformation (pre-F) via the introduction of stabilizing mutations to prevent the switch to the less immunogenic post-F conformation [30].

Subunit vaccines based on the use of stabilized pre-F protein as a vaccine antigen are currently the closest to approval for use in protecting older adults from RSV-induced LRTI. A proof-of-principle vaccine candidate, DS-Cav1 produced by researchers at the National Institute of Allergy, Immunology, and Infectious Diseases (NIAID), contains a stabilized form of the RSV pre-F protein and induced a large increase in RSV-specific neutralizing antibodies in healthy adults (18–50 years) a Phase 1 clinical trial (NCT03049488) with neutralizing antibodies present above baseline levels for at least 44 weeks [31]. Later analysis of samples collected during clinical trials conducted with DS-Cav1 and MEDI7510 (a vaccine candidate containing RSV post-F) showed that DS-Cav1 induced antibodies that more potently neutralized both RSV-A and -B strains due to targeting of antigenic sites which are only present on the pre-F protein [32▪]. Another subunit vaccine candidate (RSVPreF3) has been developed by GlaxoSmithKline Biologicals (GSK) and assessed in a Phase 2 clinical trial (NCT04126213) in pregnant women (18–40 years) [33] and a Phase 3 clinical trial (NCT04886596) in older adults (≥60 years) (34). Inoculation of pregnant women with RSVPreF3 during the late second or third trimester induced high levels of neutralizing antibodies against both RSV-A and RSV-B subtypes leading to transfer of maternal RSV-specific antibodies to infants, which persisted until at least 6 months post-birth [33]. An efficacy of 94.1% against severe RSV-induced LRTI and 71.7% against RSV-related acute respiratory infection was observed in older adults inoculated with RSVPreF3 [34▪▪]. A bivalent RSV preF vaccine candidate from Pfizer contains stabilized prefusion F glycoproteins from RSV A and B subtypes and has also been shown in a Phase 2b clinical trial (NCT04032093) to induce high levels of neutralizing antibodies against RSV-A and -B subtypes in pregnant women with concomitant placental transfer to infants [35▪▪]. This vaccine is currently being tested in a large Phase 3 clinical trial (NCT05035212) for efficacy in reducing RSV-induced LRTI in adults.

The use of virus vectors that express one or more RSV proteins is also a viable RSV immunization

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strategy. A replication incompetent human adenovirus 26 vector expressing the RSV pre-F protein (Ad26.RSV.preF) has been developed by Janssen Vaccines & Prevention and was shown to induce large increases in RSV-specific neutralizing antibodies in a small Phase 1/2a clinical trial (NCT03303625) in adults (18–50 years) and RSV-seropositive children (12–24 months) [36]. A subsequent Phase 2b clinical trial (NCT03982199) in older adults (≥65 years) inoculated with Ad26.RSV.preF and RSV preF antigen resulted in RSV A2 neutralizing antibody titers increasing by a factor of 12.1 from baseline on day 15 post-infection along with concomitant reductions in RSV-mediated LRTI [37]. The efficacy of Ad26.RSV.preF (RT-qPCR) has also been validated in the human challenge model (NCT03334695) in which healthy adults aged 18–50 years received Ad26.RSV.preF or a placebo control before intranasal challenge at 28 days with RSV-A Memphis 37b [38]. Although the young adult volunteers in this trial were not representative of groups most likely to get RSV-induced LRTI, reductions were observed in disease severity and virus copy number (via RT-qPCR assay) in conjunction with a corresponding geometric mean fold increase of 5.8 in neutralizing antibody titers against the RSV A2 strain. A second adenovirus vector-based vaccine strategy developed by GSK is based on a chimpanzee-derived replication-deficient adenovirus vector, which expresses the RSV N, F and M2–1 proteins (ChAd155-RSV) and was shown in a Phase 1 clinical trial (NCT02491463) to increase both humoral and T-cell responses in 18 to 45-year-old adults [39]. A later phase 1/2 trial (NCT02927873) in 12 to 23-month-old RSV-seropositive children showed a dose-dependent increase in RSV virus neutralizing antibodies following a first vaccination with ChAd155-RSV but not after a second vaccination [40]. Further studies are required to determine the ability of both Ad26.RSV.preF and ChAd155-RSV vaccines to protect against RSV-induced severe LRTI in susceptible groups.

An alternative viral vector has been the development by Bavarian Nordic of a recombinant modified vaccinia Ankara (MVA) strain, which expresses the RSV N (subtype A), F (subtype A), G (subtypes A, B), and M2 (subtype A) proteins (MVA-BN-RSV) [41]. This vaccine induced strong T-cell responses (1.8 to 3.8-fold increase) and a moderate increase (two-fold increase) in virus neutralizing antibodies in adults (18–49 years) and older adults (50–65 years) in a Phase 1 clinical trial [41]. A subsequent Phase 2 clinical trial (NCT02873286) in older adults (≥55 years) confirmed the increase in RSV-specific neutralizing antibodies (1.6 to 3.4-fold increase) and a strong Th1-biased cellular immune response (5.4 to 9.7-fold increase) [42▪]. The use of a booster dose of MVA-BN-RSV at 12 months post-vaccination increased the antibody and T-cell responses to a level similar to or higher than the peak initial response. This vaccine candidate is especially interesting considering recent studies highlighting the synergistic benefits for RSV immune responses of expressing both pre-F and G proteins [43,44].

Live attenuated vaccines have been proposed for use in the 6 months to 5 years of age cohort of children, a group not currently covered by other proposed RSV vaccines. This strategy has been primarily based on the long-term development by researchers at NIAID on recombinant RSV-A2 strains containing gene deletions in the presence or absence of attenuating mutations in the polymerase (L) gene. Two complicating factors have been obtaining the correct balance between attenuation and over-attenuation in vaccinated children and ensuring no reversion of introduced mutation(s) following replication of vaccine candidates in children [45]. However, continual evolution of this strategy has resulted in two candidate vaccines based on recombinant A2 (rD46) containing deletions of the NS2 gene, 112 nucleotides in the SH noncoding sequence, and codon 1313 in the L gene (conferring mild temperature sensitivity) and a stabilizing mutation in the adjacent codon (I1314L) (RSV/ΔNS2/Δ1313/I1314L) or deletion of the M2–2 gene (RSV/276) [46,47▪]. These have both been shown to be efficacious in Phase 1 clinical trials (NCT03227029 and NCT03422237) for children aged between 6 and 24 months [47▪]. Such live attenuated vaccines remain a viable route toward boosting antibody titers in young children before the onset of the annual winter epidemic of RSV infections.

A nucleic acid based vaccine, which targets all population groups susceptible to severe RSV-induced LRTI, has been developed by Moderna based on mRNAs encoding for a stabilized form of the RSV pre-F protein (mRNA-1345), which are encapsulated within lipid nanoparticles [48]. This utilizes technology successfully applied to the production of efficacious mRNA vaccines for SARS-CoV-2 during the COVID-19 pandemic. A Phase 1 clinical trial (NCT04528719) of mRNA-1345 in younger adults (18–49 years) and older adults (65–79 years) significantly boosted RSV-specific antibody titers above baseline for 6 months [49]. Further validation of this vaccine is anticipated over the next 12–18 months following publication of data from several ongoing clinical trials.

In common with efforts to better standardize RSV nomenclature and classification, the more common use of controls such as the International

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Standard for Antiserum to RSV (NIBSC code 16/284) and expression of VNT assay results in international units per milliliter (IU/ml) will make it easier to compare results from clinical trials of different RSV vaccine candidates [50]. It is also important that contemporary RSV-A (ON1) and RSV-B (BA9–11) strains are used in these assays rather than prototypic strains such as A2, Long, or CH-18537. This is supported by the recent observation that pediatric patient sera collected from patients infected with different subtypes and genotypes of RSV displayed differential antibody titers depending on which virus strain was used in the standardized VNT assays [51▪]. It is therefore important that recombinant RSVs expressing fluorescent or luciferase proteins, which are used in high throughput analysis of VNTs, are based on contemporary clinical isolates (Fig. 1) [52]. Another important factor to be considered with the design of VNT assays is that the G protein which is incorporated into virus particles generated on transformed cell lines such as Vero, MRC-5, or HEp-2 is smaller than that present in clinical samples or virus stocks produced in primary epithelial cells grown on air-liquid interface (55–95 vs. 170 kDa G protein) [53,54▪▪]. It remains to be determined how the additional N and O-linked glycosylation present in the authentic G protein (170 kDa) may impact on the accuracy of antibody titers determined in VNT assays, which use virus stocks containing smaller G proteins (55–95 kDa).

THERAPEUTICS

In addition to vaccines, a multitude of therapeutics are also likely to soon alleviate the morbidity and mortality burden caused by RSV infections. Nirsevimab (Beyfortus) is a human monoclonal antibody (mAb) developed jointly by AstraZeneca and Sanofi, which binds to antigenic site Ø in the prefusion confirmation of the RSV F protein. The extended half-life of this antibody means that just a single dose delivered via the intra-muscular route is sufficient to confer protection for a whole RSV season. A Phase 2b clinical trial (NCT02878330) and a Phase 3 MELODY clinical trial (NCT03979313) assessed the ability of Nirsevimab to protect healthy infants born preterm or at full term against RSV-induced severe LRTI [55]. Nirsevimab proved to be extremely efficacious with a relative risk reduction (RRR) of 79.5% in RSV-induced LRTI [55]. These findings have been confirmed in a Phase 3 clinical trial (NCT03979313) in which nirsevimab had an efficacy of 74.5% in preventing RSV-induced severe LRTI [56▪▪]. Nirsevimab received approval in the EU and UK in November 2022 for use in preventing RSV-induced severe LRTI in neonates and infants during their first RSV season [57]. However, the long-term success of nirsevimab will be dependent on the ongoing expansion of worldwide sequencing of clinical samples from RSV patients to enable the rapid identification of any variants containing mutations previously shown to enable virus escape from nirsevimab-mediated neutralization. This is especially relevant considering the failure of suptavumab (REGN2222), a human mAb targeted to antigenic site V, to neutralize RSV-B strains in a Phase 3 clinical trial in preterm infants (NCT02325791) due to two mutations (L172Q and S173L) in the F protein, which reduced antibody binding and neutralizing activity [58]. Thus far, only limited numbers of mutations relevant for nirsevimab efficacy have been reported including two consensus RSV-B sequences containing a single amino acid change (K68N) in the F protein, which were identified in a large sequencing study of 319 nasopharyngeal swabs collected during 2017–2020 [59▪].

Several antiviral drugs targeted to different RSV proteins are also in development but are not yet approved for use in treating RSV patients. EDP-938 is an inhibitor of RSV replication developed by Enanta Pharmaceuticals, which targets the activity of the nucleocapsid (N) protein, and has shown efficacy in a Phase 2a clinical trial (NCT03691623) using the human challenge model in which participants who had been inoculated with RSV-A Memphis 37b received either EDP-938 or a placebo [60]. EDP-938 treatment resulted in lowering of the viral load, total symptom scores, and mucus weight compared with the placebo control group in the absence of safety concerns [60]. Another antiviral that has shown promise in a clinical trial (NCT03382431) is PC786, a polymerase inhibitor (Pulmocide Ltd), which was delivered via nebulization to volunteers inoculated with RSV-A (Memphis 37b) [61]. The viral load in the PC786-treated group was lower than in the placebo group both during and after treatment, with a statistically significant reduction (32.57%) observed in viral load [61]. Additional promising antivirals in preclinical testing include 4′-fluorouridine (4′-FlU, EIDD-2749), a ribonucleoside analog capable of inhibiting polymerase functioning, which has shown efficacy in the mouse model of RSV infection [62] and the steroidal alkaloid cyclopamine and its chemical analogue A3E, which inhibits RSV infection by disorganizing and hardening inclusion body condensates in RSV-infected cells [63]. Future improvements in preclinical development of therapeutics will likely include the use of more authentic models of RSV infection such as human airway organoids [64▪▪] or culture of nasal, bronchial, or small airway organoids at

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air-liquid interface [65,66]. In the longer-term, the ability of antiviral drugs to reduce the burden of RSV-induced LRTI may be dependent on further improvements in rapid RSV diagnostics to enable quicker treatment of RSV-infected individuals in high-risk groups.

CONCLUSION

It is important to note that the recent positive developments towards efficacious RSV vaccines and therapeutics have been underpinned by long-term exploratory research on better understanding the structure and associated antigenic sites of the pre and post-F proteins. This will in future undoubtedly be complemented by an increased understanding of the function(s) of the G protein in the RSV infectious cycle, especially with respect to how the extensive and evolving N and O-linked glycosylation in the authentic form of the G protein (170 kDa) influences virus entry, shielding of antigenic sites and interactions with the host innate and adaptive immune system. The upcoming introduction of approved vaccines and antibody/antiviral therapeutics for RSV will also demand an increased worldwide focus on virus epidemiology with a particular focus on evolution of antigenic sites in the F protein, which might reduce the efficacy of vaccines or confer escape from therapeutics. This expansion in WGSs for RSV will necessitate the more routine use of recombinant viruses derived from ON1 and BA genotypes to rapidly determine the consequences of specific amino acid changes for the efficacy of currently approved vaccines and therapeutics. Similarly, the use of GMP approved virus stocks derived from contemporary circulating strains (ON1/BA genotypes) will also serve to further integrate the human challenge model into the RSV research landscape. Ultimately, the successful implementation of a new generation of RSV vaccines and therapeutics and the consequent societal benefits will depend on the continued integration and crosstalk between fundamental and translational research.

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There are no conflicts of interest.

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Papers of particular interest, published within the annual period of review, have been highlighted as:

- ■ of special interest
- ■■ of outstanding interest

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